William Gan HanSong

Cloud DevSecOps Engineer – Former Software Engineer – AWS Certified Solutions Architect – ITIL 4 Certified

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PROFILE SUMMARY

I am a **Cloud DevSecOps Engineer** with **3+ years of working experience** across **AWS & Azure Cloud Services, DevOps, and Software Engineering**. I am proficient with **Kubernetes, Docker, Terraform (IaC), Networking, Security, RBAC, CI/CD Pipelines**, Python and Bash Automation, **Logging and Monitoring Tools** with knowledge of **auto-scaling, load balancing, DNS management**, and API Testing.

PROFESSIONAL EXPERIENCE

Cloud DevSecOps Engineer (Microsoft Azure, AI Platform) @ NCS Group *Nov 2025 – Present*

- Support new application releases by **debugging Helm releases, pods and logs on Azure Kubernetes Service**, collaborating with developers to troubleshoot release issues.
- Secured a **large-scale Azure Kubernetes Service cluster with 50+ running containers** integrated with Azure SQL, Azure Event Hubs, Azure Service Bus, Azure Key Vault, and Azure Storage, using **Kubernetes network policies**.
- Developed a cloud native **end-to-end certificate management system** on **Microsoft Azure Function Apps** to create and automate certificate renewal, integrating **Microsoft Active Directory Certificate Services (ADCS) and Azure Key Vault** using **Python, Azure SDK, Azure Key Vault APIs, NTLM authentication and the ADCS Web Enrollment API**.
- Designed and integrated **end to end secrets management** between **Azure Kubernetes Service (AKS) and Azure Key Vault** using **Secrets Store CSI Driver and External Secrets Operator**, leveraging **Managed Identity, Workload Identity and Service Principal** authentication.
- Participated in **Kubernetes node scaling exercises** to validate **cluster scalability, workload resilience, pod scheduling behavior and application availability** during scale up and scale down operations.
- Provisioned Azure resources with least-privilege access **(service principals, managed identities, scoped RBAC)** and performed **system hardening** through resource-level **network settings** and **network security groups**.

Cloud Engineer (Microsoft Azure) @ Anacle Systems *July 2025 – Sep 2025*

- Performed **monthly operating system and antivirus maintenance** to ensure system reliability and security.
- Applied patches to **remediate vulnerabilities** in installed applications inside **Azure VMs**, reducing potential security risks.
- Managed internal user accounts through **Google Admin and Active Directory.**
- Conducted **Log Reviews** in compliance with SOC 2 standards, contributing to **Audit Readiness and Security Monitoring**.
- Performed **KQL queries in Log Analytics** to troubleshoot errors in the Application Gateway.
- Gained exposure to **FortiGate firewall policies** to support secure network access and compliance requirements.

Pursued Degree and Various Technical Work *Jan 2023 – Feb 2025*

- **Completed further studies** - **BSc in Information Technology, scoring distinction in all modules**, graduated on **18 Feb 2025.**
- Explored **Infrastructure as Code (IaC)** using Terraform to fully **automate provisioning and enable replicable, consistent, and efficient deployment**.
- Explored **Ansible** to fully automate configuration, setting up a **highly available Kubernetes cluster with HAProxy and Keepalived**.

Software Engineer @ NCS Group *Aug 2021 – Oct 2022*

- **Improved job execution time by 20%** through optimizing and upgrading **python-based data processing pipelines**.
- Fulfilled **data extraction and reporting requests** from stakeholders by **writing SQL queries for Microsoft SQL Server**.
- Remediate **OWASP vulnerabilities** by **patching dependencies** across internal microservices.

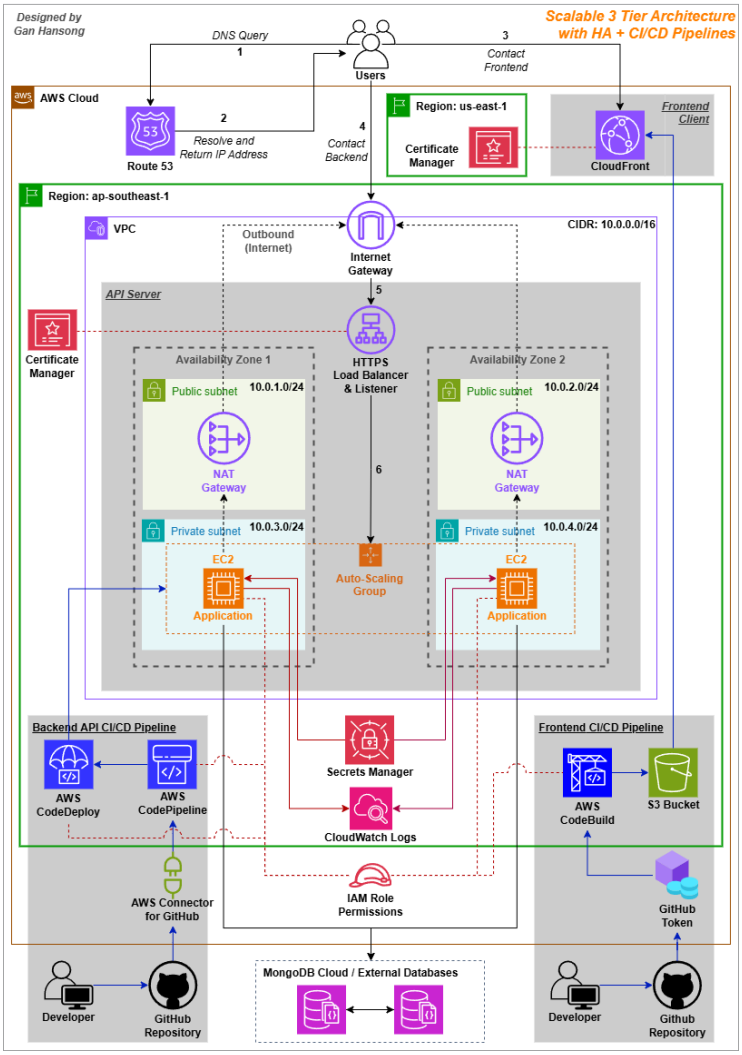
EDUCATION

- **BSc in Information Technology, First Class Honours** – Teesside University **Completed in Feb 2025**
  - ❖ **Modules:** Information Governance, Service Management, Computing Project, Databases and Business Intelligence, Client Focused Business Solutions
- **Diploma in Electrical and Electronics Engineering** – Republic Polytechnic Completed in May 2018

NOTABLE CERTIFICATIONS

- HashiCorp Certified: Terraform Associate – Issued by **IBM** (Infrastructure as Code, Automation) **Acquired in Oct 2026**
- AWS Solutions Architect Associate – Issued by **AWS** (AWS Cloud, System Architecture) **Acquired in May 2024**
- **Google Data Analytics Professional** – Issued by **Google** (SQL, R Programming, Tableau, Excel) **Acquired in Jun 2021**
- **ITIL 4 Foundation Certificate** – Issued by **PeopleCert** (IT Service Management) **Acquired in May 2025**
- **Meta Front-End Developer** – Issued by **Meta** (React, UI/UX Design, Requirements Gathering) **Acquired in Aug 2023**

My Personal Projects



Scalable 3-Tier Architecture featuring AWS, CDN, High Availability & CI/CD Pipelines

The application is deployed in a scalable and high-availability system leveraging AWS Cloud services and deployed using Terraform (Infrastructure as Code) tool.

Frontend Application: Hosted static assets on S3 with global distribution through CloudFront (CDN). Built with React, Redux, Node.js.

Frontend CI/CD Pipeline: Used AWS CodeBuild to detect changes from GitHub, pull the repository and build, and store in S3 bucket, then read by cloudfront.

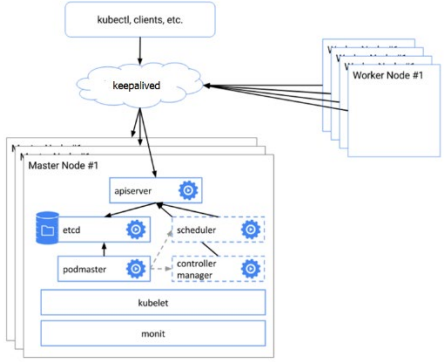
Backend Application: Used EC2 instances in Auto Scaling Groups behind a Load Balancer for scalability. Built with Express.js, MongoDB, Node.js and JWT Authentication.

Backend CI/CD Pipeline: CodePipeline to detect changes on GitHub, trigger CodeDeploy to release new versions onto EC2 instances in the Auto Scaling Group with zero downtime using Blue/Green Deployment.

Services Used: Route 53, Certificate Manager, Secrets Manager, IAM, S3, CloudFront, CloudWatch Logs, VPC, Nat Gateway, IGW, Subnets, Route Tables, EC2, Load Balancer, Target Groups, Auto Scaling Groups, CodeBuild, CodePipeline, CodeDeploy.

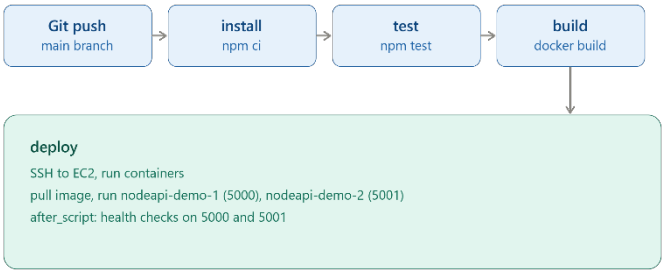
Frontend (App): https://github.com/hswg94/ecomm-react-frontend
Backend (App): https://github.com/hswg94/ecomm-express-api/
Terraform (IaC): https://github.com/hswg94/ecomm-terraform
Project Documentation: https://github.com/hswg94/ecomm-terraform/blob/main/Project%20Documentation.pdf
Live Demo: https://ecomm.hswg94.com

Highly Available K8s Cluster using HAProxy and Keepalived (Deployed with Ansible)



https://github.com/hswg94/ansible-k8s-ha-cluster

Simple GitLab CI/CD Demo



https://github.com/hswg94/gitlab-cicd-demo